

Cold Chain Shrinkage Analysis for Company X

Role: Data Analyst

Focus: Stock loss detection, supplier performance, warehouse control, and operational risk reduction

Project Summary

Company X needed a clearer picture of where inventory was being lost across its cold chain operations. The business problem was not simply “what is missing,” but **where it is missing, why it is missing, and which parts of the supply chain are driving the loss.**

This analysis examined the full product journey, from supplier purchase orders to warehouse transfers, storage conditions, and stock write-offs. The goal was to identify the main drivers of shrinkage, isolate the weakest control points, and provide practical actions that could help Company X protect stock value and strengthen operational control.

Business Objective

The analysis was built to answer four core questions:

1. What is going missing?
2. Where is it going missing?
3. Why is it going missing?
4. Which supplier, warehouse, or product line should be addressed first?

The intended audience for this work includes **operations leaders, procurement teams, warehouse managers, and supply chain decision-makers.**

Data Scope

The dataset contained seven key sheets:

- **suppliers_coldchain** — supplier details, lead times, and supplier metadata

- **products_coldchain** — product attributes such as shelf life, supplier, and storage temperature requirements
- **purchase_orders_coldchain** — purchase order tracking and order status
- **purchase_order_lines_coldchain** — line-level purchasing transactions
- **stock_movements_coldchain** — inventory movement across the cold chain network
- **cycle_counts_coldchain** — system and physical inventory counts
- **temperature_logs_coldchain** — warehouse temperature and humidity records

Each sheet represented a different stage of the inventory lifecycle and helped build a complete view of how goods moved through the business.

Data Preparation

A detailed cleaning and validation process was carried out before analysis.

Supplier, Product, and Purchase Order Data

- Duplicate records were removed from the supplier, product, and purchase order sheets.
- Text fields were standardized using proper text formatting to improve consistency and usability.

Purchase Order Lines

This sheet was validated more deeply through calculated checks.

- Two validation columns were added:
 - **line_total_actual**
 - **line_total_check**

These were used to recalculate expected totals and compare them against the recorded values. This process flagged **6 line total errors**, improving confidence in the accuracy of purchasing data.

Stock Movements

This sheet was also rebuilt for traceability.

- Duplicate records were removed.
- Two validation columns were added:
 - **missing_reference_id_check**
 - **reference_id_actual**

These checks recovered **4 missing reference IDs**, which improved data completeness and tracking integrity.

Cycle Counts

- Duplicate records were removed.
- Two validation columns were added:
 - **system_qty_check**
 - **variance_check**

Both checks confirmed that:

- **100% of system quantity values passed validation**
- **100% of variance values passed validation**

This indicated strong reliability in the inventory count data.

Analytical Approach

After cleaning, the dataset was moved into **SQL** for structured analysis and reporting. Four reporting tables were created to answer the business questions more effectively:

Purchase_Order_Tracking_Sheet

Tracks products from supplier order through to Company X receipt.

A **delivery_status** field was added to assess supplier fulfillment and highlight where the largest losses were happening.

Warehouse_Transfer_Log

Tracks product movement between Company X warehouses.

A **shrinkage_check** field was added to determine whether inventory loss occurred during internal transfers.

Warehouse_Writeoff_Summary

Tracks losses caused by damage or temperature excursion while stock is in storage or cold rooms.

Warehouse_Health_Status_Report

Monitors warehouse operational health and identifies risk areas such as offline sensors or inconsistent environmental control.

This structure made it possible to separate inventory loss into distinct stages and identify where the business was most exposed.

Key Findings

1. The Main Loss Point Is Before or During Receipt From Suppliers

Over the three-month period from **January to March**, Company X placed **more than 420 orders** for different products.

- **Total units ordered:** 53,957
- **Total units received:** 36,198

This represents approximately **33% shrinkage**.

That level of loss is significant. It suggests the largest issue is occurring **before goods fully enter Company X's control**, most likely during supplier fulfillment, transport, unloading, or receipt verification.

Possible causes include:

- Supplier shortages
- Transit loss
- Damaged packaging
- Theft
- Counting errors

This is the most urgent loss area and the one with the greatest financial impact.

2. Supplier Performance Is Uneven

A closer look at suppliers showed large differences in fulfillment quality.

Most used supplier: SUP00013

- Units ordered: **2,237**
- Units received: **1,619**
- Fulfillment rate: approximately **72%**

Although this supplier performed better than the overall average, the result still points to a meaningful level of loss.

Top 10 suppliers combined

- Units ordered: **13,404**
- Units received: **8,951**
- Fulfillment rate: approximately **67%**

This shows that even the suppliers used most often are contributing heavily to shrinkage.

3. Supplier Fulfillment Benchmarks

Using the grading scale below:

- **A+** = 96–100%
- **A** = 90–95%
- **B** = 80–89%
- **C** = 70–79%
- **D** = 60–69%
- **E** = 50–59%
- **F** = Below 50%

The supplier picture breaks down as follows:

- **Overall supplier fulfillment:** ~67% → **D / Fair-Pass**
- **Most used supplier (SUP00013):** ~72% → **C / Good**
- **Best-performing supplier group:** ~92% → **A / Excellent**

Highest fulfillment rate

- **Supplier ID:** SUP00027
- Units ordered: **437**
- Units received: **417**
- Fulfillment rate: approximately **95%**
- Grade: **A / Excellent**

Lowest fulfillment rate

- **Supplier ID:** SUP00120
- Units ordered: **178**
- Units received: **0**
- Fulfillment rate: **0%**
- Grade: **F / Fail**

Least used supplier

- **Supplier ID:** SUP00009
- Units ordered: **53**
- Units received: **45**
- Fulfillment rate: approximately **85%**
- Grade: **B / Very Good**

This confirms that supplier reliability is not uniform and should be managed more actively.

4. Internal Warehouse Transfers Are Not the Problem

The internal transfer analysis was clear:

0 shrinkage occurred during warehouse-to-warehouse movement.

This is an important control insight because it removes internal transfer as a major source of loss and shifts attention back to supplier-side delivery and receiving.

5. Storage Loss Is Minimal, But Not Zero

Damaged packaging / tampering

Across the 10 warehouses, only **44 units** were written off due to damaged packaging.

Given that Company X received **36,198 units**, this represents **less than 1% shrinkage**.

Temperature excursion

Across the same warehouses, only **36 units** were lost to temperature excursion.

This also represents **less than 1% shrinkage**.

These results suggest storage controls are generally strong, but they still need continuous oversight.

6. Warehouse Health Needs Attention

Although loss from storage conditions was low, warehouse monitoring showed a risk issue:

- **WH-PHC-01**
- **WH-IBA-01**

- **WH-BEN-01**

These three warehouses had sensors go offline.

That matters because warehouse sensors are critical for:

- temperature monitoring
- cold room compliance
- product quality protection
- early detection of equipment failure

Even when losses are not yet visible, offline sensors create exposure and should be treated as an operational warning sign.

Business Interpretation

The analysis shows that Company X's shrinkage problem is **not primarily happening inside the warehouse network**. Instead, the biggest losses are occurring **upstream**, before goods are fully received into the business.

That means the highest-value improvements will come from:

- tighter receiving controls
 - better supplier oversight
 - stronger chain-of-custody accountability
 - clearer separation of shrinkage causes
 - regular performance monitoring
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Recommendations

1. Establish a Receiving Control System

The company needs a formal receiving process that records exactly how much stock arrives, in what condition, and whether it matches the order.

Without this, it is impossible to know whether losses came from:

- supplier shortages
- transit loss
- damage
- theft

- counting errors

A receiving control system gives the business its first reliable checkpoint and makes shrinkage measurable.

2. Introduce Supplier Performance Tracking

Supplier quality varies significantly. A dedicated supplier performance report should monitor:

- fulfillment rate
- delivery accuracy
- recurring shortages
- shipment quality

This would allow Company X to identify poor-performing suppliers early and make better sourcing decisions.

3. Improve Chain of Custody

Every handover should have a clear owner.

A proper chain of custody should follow:

Supplier → Transporter → Warehouse Receiver → Inventory Team

This creates accountability at every stage and makes it easier to locate where loss occurs.

4. Seal and Track Shipments

Large shipments should be sealed and tracked so the business can confirm whether goods were opened, altered, or tampered with during transit.

5. Separate Types of Shrinkage

Not all losses come from the same cause.

Supplier shortage, transport loss, warehouse damage, and internal theft should be tracked separately so that corrective action matches the actual issue.

6. Perform Monthly Root Cause Analysis

Shrinkage should not only be recorded; it should be explained.

A monthly root cause review will help Company X:

- identify repeating patterns
- validate whether corrective actions are working

- monitor supplier and warehouse trends
 - improve operational control over time
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Conclusion

This project showed that Company X has a serious shrinkage challenge, with the largest losses happening **before or during receipt from suppliers**. At the same time, internal warehouse transfers and storage-related losses are relatively well controlled.

The most important outcome of the analysis is this:

Company X does not need a generic fix. It needs a targeted control strategy focused on receiving, supplier accountability, and chain-of-custody visibility.

By strengthening those areas, the company can reduce stock loss, improve operational discipline, and protect inventory value across the cold chain.